

Subject Description Form

Subject Code	COMP2014							
Subject Title	Mathematics for Machine Learning							
Credit Value	3							
Level	2							
Pre-requisite / Co-requisite / Exclusion	NIL							
Objectives	The objectives of this subject are to: • provide the basic concepts of mathematics for machine learning • apply practical problem-solving skills by using appropriate numerical computing environments and mathematical software libraries							
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Professional/academic knowledge and skills</u> (a) understand the basic concepts of mathematics for machine learning; (b) apply proper mathematics concepts in solving problems; (c) familiarize with numerical computing environments and mathematical software libraries; <u>Attributes for all-roundedness</u> (d) solve problems independently; and (e) think critically for improvement in solutions.							
Subject Synopsis/ Indicative Syllabus	<table border="1"> <thead> <tr> <th>Topic</th><th>Duration of Lectures</th></tr> </thead> <tbody> <tr> <td> 1. Numerical computing software Mathematical software libraries (e.g., Numpy, Pandas, Scikit-Learn; numerical computing environments (e.g., Pytorch, Keras, Matlab) </td><td>4</td></tr> <tr> <td> 2. Linear algebra Vector space; linear transformation and matrices; solving linear equations; matrix decomposition; eigenvalues and singular values; matrix norms and condition numbers; inverse matrices; column space and null space; non-square matrices as transformation between dimensions; pseudo-inverses </td><td>12</td></tr> </tbody> </table>		Topic	Duration of Lectures	1. Numerical computing software Mathematical software libraries (e.g., Numpy, Pandas, Scikit-Learn; numerical computing environments (e.g., Pytorch, Keras, Matlab)	4	2. Linear algebra Vector space; linear transformation and matrices; solving linear equations; matrix decomposition; eigenvalues and singular values; matrix norms and condition numbers; inverse matrices; column space and null space; non-square matrices as transformation between dimensions; pseudo-inverses	12
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	3. Optimization One-dimensional optimization; least squares; constrained optimization; convex optimization; gradient descent; stochastic gradient descent	4					
	4. Probability and Statistics Recap of basic concepts in probability; maximum likelihood estimation; Bayes' rule; statistical inference	4					
	5. Applications of mathematical skills in machine learning Linking mathematical skills with machine learning problems; examples of applications of mathematical skills in machine learning; the relevance to the machine learning subjects in COMP	2					
	Total	26					
Teaching/ Learning Methodology	Lectures provide students with the main concepts of the topic, together with comprehensive examples for easy understanding. Tutorial and lab sessions offer an opportunity to students for practicing their skills. Assignments will be used in the course to develop theoretical skills and practical skills.						
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
			a	b	c	d	e
	Continuous Assessment						
	Assignments and Quizzes	30%	✓	✓	✓	✓	✓
	Examination	70%	✓	✓		✓	✓
	Total	100%					
	Both assignments and quizzes are mainly used to assess the understanding of mathematics concepts and their applications to machine learning problems, i.e., the outcomes (a) and (b). Quizzes are expected to be short and conducted in class. In contrast, assignments may cover both short questions and long questions. Some assignment(s) can be used in numerical computing environments, i.e., the outcome (c). All assessment tasks are relevant to outcomes (d) and (e), i.e., independent problem-solving and critical thinking for improvements.						

Student Study Effort Expected	Class contact:	
	▪ Lecture	26 Hrs.
	▪ Tutorial/Lab	13 Hrs.
	Other student study effort:	
	▪ Assignments, Quizzes, Self-study	66 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	Reference Books: 1. Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong: Mathematics for Machine Learning, Cambridge University Press, 2020. PDF: https://mml-book.github.io/ 2. Trevor Hastie, Jerome H. Friedman, Robert Tibshirani: The Elements of Statistical Learning: Data Mining, Inference, and Prediction. Second Edition, Springer Series in Statistics, Springer 2001. PDF: https://hastie.su.domains/Papers/ESLII.pdf 3. Ian Goodfellow and Yoshua Bengio and Aaron Courville: Deep Learning, MIT Press, 2016. 4. Christopher Bishop: Pattern Recognition and Machine Learning, 5th Edition. Information science and statistics, Springer 2007 5. A. Quateroni, F. Saleri, and P. Gervasio: Scientific Computing with MATLAB (Texts in Computational Science and Engineering, 2), 4th Edition, Springer 2014.	